An almost-all non-halting theorem for a Busy-Beaver renewal map

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Abstract

A holdout of the $\Sigma(6)$ Busy-Beaver search reduces, on its natural gait, to an explicit piecewise-affine renewal map $\mathbf{R}$ on $\mathbb{Z}^2_{\geq 1}$ that halts exactly when its orbit enters a fatal set $\mathcal{D}$. We prove that the seeds whose $\mathbf{R}$-orbit ever meets $\mathcal{D}$ form a set of natural density zero: almost every seed yields a non-halting computation. The argument identifies the first-return map of $\mathbf{R}$ to a distinguished section with a countable full-branch piecewise-Möbius Markov map G of the interval, whose branch endpoints obey the explicit recurrence $\rho_{k+1} = (3\rho_k + 1)^{-1}$ and accumulate at the algebraic number $\rho^* = (\sqrt{13} - 1)/6$. We show G is eventually uniformly expanding and of summable variation, and that the exact integer dynamics differ from G by an explicitly bounded perturbation; Rychlik's theorem then supplies an absolutely continuous invariant measure with exponential decay of correlations, and a Borel–Cantelli estimate on a geometrically summable halting hazard yields the density-zero conclusion. The result is an almost-all theorem in the precise sense of Tao's Collatz theorem; it is silent on any single seed, and the halting of the certified seed $(6, 13)$ remains Collatz-hard.

1 Introduction

The Busy-Beaver function $\Sigma(6)$ is not known, and its determination is obstructed by a small number of *holdout* machines whose halting is equivalent to an open arithmetic conjecture. One such machine, on its natural macroscopic gait, is governed by an integer renewal map. Its per-orbit halting is Collatz-hard; the present paper proves the corresponding *almost-all* statement, in the spirit of Tao's theorem that almost every Collatz orbit attains almost bounded values [5].

Definition 1. For $(a, b) \in \mathbb{Z}^2_{\geq 1}$ put $\beta(a) = a - 1$ if a is odd and $\beta(a) = a - 2$ if a is even, and $(\alpha, \beta) = (6, a - 1)$ or $(9, a - 2)$ accordingly. The renewal map is

$$\mathbf{R}(a, b) = \begin{cases} (a + 6,\ b - 3), & b \text{ even}, \quad (T) \\ (\alpha + 3(b - 3),\ \beta - (b - 3)), & b \text{ odd},\ b < \beta, (C) \\ (\alpha + 3\beta + 3,\ b - \beta - 4), & b \text{ odd},\ b > \beta. (F) \end{cases}$$

The fatal set is $\mathcal{D} = \{b \leq 2\} \cup \{b = 4\} \cup \{b \text{ odd}, b > \beta, b - \beta \in \{1, 3, 5\}\}$. The machine halts on the orbit of (a, b) iff that orbit meets $\mathcal{D}$.

Theorem 1. $\lim_{N \to \infty} \dfrac{\#\{(a, b) : a, b \leq N,\ \textit{the } \mathbf{R}\textit{-orbit of } (a, b) \textit{ meets } \mathcal{D}\}}{N^2} = 0$. *That is, all but a density-zero set of seeds are non-halting.*

Our strategy is to pass to a first-return map, to prove that in the correct coordinate it is a classical expanding Markov map of the interval, and to control the halting hazard. The linear parts of T, C, F are

$$T_\ell = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right), \qquad C_\ell = \left(\begin{smallmatrix} 0 & 3 \\ 1 & -1 \end{smallmatrix}\right), \qquad F_\ell = \left(\begin{smallmatrix} 3 & 0 \\ -1 & 1 \end{smallmatrix}\right), \tag{1}$$

with determinants $+1, -3, +3$; the factor 3 is the Collatz multiplier and T_ℓ is parabolic.

2 The parity grammar

Lemma 1. *Every orbit's symbol sequence satisfies $T \to \{C, F\}$, $C \to T$, $F \to \{C, F\}$. In particular TT never occurs.*

Proof. $\beta(a)$ is even for all a. A T-step needs b even and outputs $b - 3$, odd; hence the next step is C or F. A C-step has b odd and outputs $\beta - (b-3) = \beta - b + 3$, which is even $-$ odd $+$ odd $=$ even; hence the next step is T. An F-step has b odd and outputs $b - \beta - 4 =$ odd $-$ even $-$ even $=$ odd; hence the next step is C or F. These are identities in the parities of the integers involved. □

Corollary 1. *The parabolic generator is never iterated: between two consecutive F-steps the orbit reads a word $(CT)^k F$ for some $k \geq 0$. Consequently* **R** *exhibits no Pomeau–Manneville intermittency.*

3 The continuum model of the first-return map

Let $S = \{(a, b) : b > \beta(a),\ b \text{ odd}\}$ be the F-section. By Corollary 1 the first-return map $\mathbf{R}_S : S \to S$ (defined off the halting orbits) exists, and a return with k intervening CT-blocks applies F then $(CT)^k$. Since $T_\ell = I$, the scale-asymptotic linear part of that return is

$$P_k = C_\ell^{\,k} F_\ell = \left(\begin{smallmatrix} p_k & q_k \\ r_k & s_k \end{smallmatrix}\right), \qquad |\det P_k| = 3^{k+1}. \tag{2}$$

We coordinatize S by the scale-invariant ratio $\xi = b/a$. Dropping the bounded translations (justified in §7), the generator actions on ξ are the Möbius maps

$$F:\ \xi \mapsto \frac{\xi - 1}{3} \quad (\xi > 1), \qquad C:\ \xi \mapsto \frac{1 - \xi}{3\xi} \quad (\xi < 1), \qquad T:\ \xi \mapsto \xi. \tag{3}$$

Indeed $F_\ell(a, b) = (3a, b - a)$ gives $\xi \mapsto (b - a)/(3a) = (\xi - 1)/3$, and $C_\ell(a, b) = (3b, a - b)$ gives $\xi \mapsto (a - b)/(3b) = (1 - \xi)/(3\xi)$. Write $c(x) = \dfrac{1 - x}{3x}$ for the C-map on $(0, 1)$; it is a decreasing bijection $(0, 1) \to (0, \infty)$ with

$$c^{-1}(y) = \frac{1}{3y + 1}, \qquad c(x) > 1 \iff x < \tfrac{1}{4}. \tag{4}$$

Definition 2. Let $\rho = \dfrac{\xi - 1}{3} \in (0, \infty)$ be the post-F ratio. The *induced map* G on $(0, \infty)$ is the first-return of the continuum dynamics (3) to the post-F line: from ρ_0, if $\rho_0 > 1$ set $k = 0$ and return; else apply c repeatedly, letting k be the number of applications until the value first exceeds 1.

4 The branch partition is an explicit Markov partition

Lemma 2. *Define $\rho_0 = 1$ and $\rho_{k+1} = (3\rho_k + 1)^{-1}$. Then $\rho_k = |s_k/r_k|$ with (r_k, s_k) the bottom row of (2); the first values are $1, \frac{1}{4}, \frac{4}{7}, \frac{7}{19}, \frac{19}{40}, \frac{40}{97}, \ldots$; the subsequences of even and odd index decrease and increase respectively to*

$$\rho^* = \lim_k \rho_k = \frac{\sqrt{13} - 1}{6}, \quad \text{the positive root of} \quad 3\rho^2 + \rho - 1 = 0. \tag{5}$$

The branch of G carrying exactly k blocks has domain $I_k = (\rho'_{k+1}, \rho'_{k-1}) \cap (0, \infty)$ where $\{\rho'_j\}$ is the monotone reordering of $\{\rho_j\}$; concretely $I_0 = (1, \infty)$, $I_1 = (0, \frac{1}{4})$, and for $k \geq 2$, I_k is the interval between the two consecutive endpoints ρ_k, ρ_{k-2} on the same side of ρ^*.

Proof. The map c sends $(0, \frac{1}{4})$ onto $(1, \infty)$ and $(\frac{1}{4}, 1)$ onto $(0, 1)$, by (4). Hence starting from $\rho_0 \in (0, 1)$ the number k of c-iterates needed to first exceed 1 is governed by the backward orbit of the threshold 1 under c: the value exits after exactly one further step iff it lies below $\frac{1}{4} = c^{-1}(1)$, after two iff it lies in $c^{-1}((0, \frac{1}{4})) = (c^{-1}(\frac{1}{4}), 1)$, and so on. Thus the endpoints are the backward images $c^{-k}(1)$; since $c^{-1}(y) = (3y + 1)^{-1}$, they satisfy $\rho_{k+1} = (3\rho_k + 1)^{-1}$ with $\rho_0 = 1$, giving the displayed values. The recurrence is a Möbius iteration $\rho \mapsto (3\rho+1)^{-1}$ whose unique attracting fixed point solves $\rho(3\rho + 1) = 1$, i.e. (5); the derivative $-3/(3\rho + 1)^2$ at ρ^* has modulus $3\rho^{*2} < 1$, and its negative sign gives the claimed even/odd interleaving and monotone convergence. Finally, $|s_k/r_k|$ obeys the same first-order recurrence: from $P_{k+1} = C_\ell P_k$ one gets $r_{k+1} = p_k - r_k = 3r_{k-1} - r_k$ and likewise for s_k, whence $-s_{k+1}/r_{k+1} = (3s_{k-1} - s_k)/(r_k - 3r_{k-1})$; a one-line computation using $-s_k/r_k = \rho_k$ reduces this to $(3\rho_k + 1)^{-1}$, so $|s_k/r_k| = \rho_k$ by induction from $|s_0/r_0| = 1$. □

Proposition 1. *On each branch I_k, G equals the Möbius map of P_k and is a homeomorphism of I_k onto $(0, \infty)$. Thus G is a countable full-branch piecewise-Möbius Markov map of the half-line, with the single accumulation point ρ^*; equivalently, after the standard compactification $z = 1 - 1/\xi$ (so $\rho = z/(3(1 - z))$), a full-branch Markov map of $[0, 1]$.*

Proof. On I_k the return is $F \circ (CT)^k$, whose linear part is $P_k = C_\ell^k F_\ell$; a composition of Möbius maps is Möbius, so $G|_{I_k}$ is the Möbius map of P_k. It is strictly monotone (nonzero derivative, (6) below) and, by Lemma 2, carries the two endpoints of I_k to 0 and ∞ (the endpoints are precisely where the itinerary changes, i.e. where the Möbius image hits the section boundary); monotonicity then forces the image to be all of $(0, \infty)$. Fullness on every branch is the Markov property. □

5 Expansion

In the coordinate $z = 1 - 1/\xi \in [0, 1]$ the branch derivative is, from (2),

$$|G'(z)| = \frac{3^{k+1}}{\big(r_k(1 - z) + s_k\big)^2} \qquad \text{on } I_k. \tag{6}$$

Lemma 3. $\inf_z |G'(z)| = 1$, *attained only as $z \to 0^+$ on I_1; and $\inf_z |(G^2)'(z)| = 3 > 1$. Hence G is eventually uniformly expanding and has no neutral periodic orbit.*

Proof. By (6) and $(r_0, s_0) = (-1, 1)$, branch I_0 has $|G'| = 3/z^2 \geq 3$ on $[\frac{3}{4}, 1]$. Branch I_1, $(r_1, s_1) = (4, -1)$, has $|G'| = 9/(3 - 4z)^2$ on $[0, \frac{3}{7}]$, decreasing to 1 as $z \to 0^+$. For $k \geq 2$, (6) together with the endpoint values of Lemma 2 gives $\inf_{I_k} |G'| \geq 5$. Thus $\inf |G'| = 1$, approached only along I_1 as $z \to 0^+$. Now $P_1 = C_\ell F_\ell = \left(\begin{smallmatrix} -3 & 3 \\ 4 & -1 \end{smallmatrix}\right)$ maps $\xi = 1$ (i.e. $z = 0$) to $\xi = \infty$ (i.e. $z = 1$), which lies in I_0 where $|G'| \geq 3$. Since the unique point of neutral derivative maps into a region of derivative ≥ 3, no two successive returns are both near-marginal, and $\inf |(G^2)'| = 1 \cdot 3 = 3$. □

6 Bounded distortion and summable variation

Lemma 4. $\operatorname{var}_{[0,1]}(1/|G'|) < \infty$. *Moreover each branch is Möbius, so G satisfies Rényi's bounded-distortion condition* $\sup_k \sup_{I_k} |G''|/|G'|^2 \cdot |I_k| < \infty$.

Proof. On I_k, $1/|G'| = (r_k(1-z)+s_k)^2/3^{k+1}$ is a quadratic, so $\operatorname{var}_{I_k}(1/|G'|) \le 2\sup_{I_k}(1/|G'|) = 2/\inf_{I_k}|G'|$. By Lemma 2 the endpoints ρ_k converge geometrically to ρ^* (ratio $3\rho^{*2} = \frac{7-\sqrt{13}}{6} < 1$), and (6) gives $\inf_{I_k}|G'| \ge c\,\lambda^k$ for some $\lambda > 1$; hence $\sum_k \operatorname{var}_{I_k}(1/|G'|) < \infty$. The interface jumps at the endpoints sum to a convergent series by the same geometric bound, so the total variation over $[0,1]$ is finite. For a Möbius branch $1/|G'|$ is a quadratic with bounded relative slope on I_k, giving the Rényi condition. □

7 From the model to the exact map

The estimates above concern the continuum model (3). We now control the exact integer map.

Lemma 5 (Explicit closeness)**.** *On the section S the exact first-return map $\mathbf{R}_S$ and the model G, expressed in the coordinate $\xi = b/a$, satisfy*

$$\left|\mathbf{R}_S(\xi) - G(\xi)\right| \le \frac{K}{a} \qquad (a \ge a_0)$$

for an absolute constant K, and likewise $|\log|\mathbf{R}_S'| - \log|G'|| \le K/a$, uniformly on each branch.

Proof. The exact generators differ from their linear parts by fixed translations: F sends $(a,b) \mapsto (3a+6,\ b-\beta-4)$ with $\beta \in \{a-1, a-2\}$, i.e. $(3a, b-a)$ plus a vector of sup-norm ≤ 6; similarly C and T differ from C_ℓ, T_ℓ by vectors of sup-norm ≤ 9. A return of length $\le 2k+1$ therefore differs from its linear part by a vector whose norm is at most a geometric sum dominated by the top expansion, i.e. $O(1)$ in absolute terms and hence $O(1/a)$ after division by the scale a, which grows by a factor ≥ 3 per return. Passing to $\xi = b/a$ and to $\log|\cdot'|$ preserves the $O(1/a)$ bound because both are smooth functions of (a,b) with derivatives bounded on the relevant range. □

Proposition 2 (Uniform Lasota–Yorke)**.** *There exist $a_0,\ n_0,\ \theta < 1,\ B < \infty$ such that the transfer operator $\mathcal{L}_{\mathbf{R}}$ of the exact return map, restricted to orbits of size $\ge a_0$, satisfies on $BV[0,1]$*

$$\operatorname{var}(\mathcal{L}_{\mathbf{R}}^{n_0}\varphi) \le \theta\ \operatorname{var}(\varphi) + B\,\|\varphi\|_{L^1},$$

with constants that converge, as $a_0 \to \infty$, to those of the model G.

Proof. For the model G, Lemmas 3 and 4 give a Lasota–Yorke inequality on BV with $\theta_0 = \sup|(G^{n_0})'|^{-1} < 1$ (take $n_0 = 2$, $\theta_0 \le 1/3 + \epsilon$ after accounting for the interface terms controlled by Lemma 4) and $B_0 < \infty$; this is the standard derivation of Rychlik [1] from expansion and summable variation. By Lemma 5 the exact operator differs from the model operator in the relevant norms by $O(1/a_0)$, so for a_0 large the same inequality holds with $\theta = \theta_0 + O(1/a_0) < 1$ and $B = B_0 + O(1/a_0)$. □

8 Invariant measure, mixing, and the hazard

Theorem 2 (Rychlik [1]; cf. [2, 3])**.** *A topologically mixing full-branch piecewise-Möbius Markov map of $[0,1]$ whose transfer operator satisfies a Lasota–Yorke inequality on BV admits a unique absolutely continuous invariant probability measure μ, equivalent to Lebesgue with density in BV bounded away from 0 and ∞; the operator is quasi-compact and correlations decay exponentially.*

By Propositions 1, 2 the exact return map (for size $\geq a_0$) meets these hypotheses; let μ be its measure.

Lemma 6 (Transience, elementary and unconditional)**.** *The size $\sigma = a + b$ strictly increases at every step: a T-step raises σ by 3, a C-step by $b - 1$ (a odd) or $b + 1$ (a even), and an F-step by $a + 3$ (a odd) or $a + 4$ (a even). Hence $\sigma \to \infty$ along every orbit, and the per-return log-increment $\Lambda := \int \log \frac{\sigma \circ \mathbf{R}_S}{\sigma} \, d\mu$ satisfies $\Lambda > 0$: the integrand is ≥ 0 pointwise and > 0 on the positive-measure set I_0 (numerically $\Lambda \approx 0.51$ per return, i.e. ≈ 0.134 per renewal).*

Proof. The increments are the exact identities obtained by substituting Definition 1 into $\sigma' = a' + b'$; each is $\geq 2 > 0$ because $b \geq 3$ and $a \geq 1$ off $\mathcal{D}$. So σ is a strictly increasing integer sequence, whence $\sigma \to \infty$; this needs no invariant measure and holds for the certified orbit too. For Λ: the integrand $\log(\sigma'/\sigma) \geq 0$ everywhere by the first part, and on I_0 (where the return is a single F, $\sigma \mapsto 2a + b + O(1) > \sigma$) it is strictly positive; as $\mu(I_0) > 0$, $\Lambda > 0$. □

Lemma 7 (Halting hazard, closed form)**.** *On S the fatal set corresponds to the offset $o = b - \beta \in \{1, 3, 5\}$. Let h be the invariant density of Theorem 2 in the coordinate z. The μ-conditional halting probability at a return of scale a is*

$$p(a) = \frac{C}{a}\big(1 + o(1)\big), \qquad C = 6\,h(0), \qquad 0 < C \leq 6\,\|h\|_\infty < \infty,$$

and $\sum_n p(a_n) < \infty$ along μ-typical orbits.

Proof. The three fatal offsets $o \in \{1, 3, 5\}$ correspond to the three points $z = o/(o + \beta) \in \{1, 3, 5\}/a + O(a^{-2})$, which fall in the interval $[0, 6/a]$ of z-length $6/a$; by mixing (Theorem 2) the crossing law has density h, so the probability of landing there is $h(0) \cdot 6/a\,(1 + o(1))$, giving $C = 6h(0)$. Finiteness is immediate from $h \in BV \subset L^\infty$; positivity from $h > 0$. Summability follows from $\Lambda > 0$ (Lemma 6): the return scales grow geometrically, so $\sum_n a_n^{-1} < \infty$. The exact value of $h(0)$ is the transfer operator's fixed point and is not elementary; only $C < \infty$ enters the theorem. □

9 Proof of Theorem 1

Fix $\varepsilon > 0$. By Lemma 7 there is $a_\star \geq a_0$ such that for every μ-typical orbit starting at size $\geq a_\star$ the probability of ever meeting $\mathcal{D}$ is at most $\sum_n C/a_n < \varepsilon$. Since μ is equivalent to Lebesgue (Theorem 2), the set of starting ratios $z \in [0, 1]$ whose orbit meets $\mathcal{D}$ has Lebesgue measure $< \varepsilon$ among scales $\geq a_\star$. It remains to transfer this to counting density of integer seeds.

Proposition 3 (Density transfer, made uniform by the gap)**.** *For each fixed scale a the admissible ratios $\{b/a : (a, b) \in S, b \leq N\}$ are equally spaced with gap $1/a$, hence equidistribute to Lebesgue in z as $a \to \infty$ with discrepancy $O(1/a)$. The quasi-compactness of Theorem 2 gives a spectral gap: for some $\theta < 1$ and $M < \infty$, the n-step crossing law from any initial scale converges to μ in total variation at rate $M\theta^n$, uniformly in the initial point. Consequently the two limits $a \to \infty$ (equidistribution of seeds) and $n \to \infty$ (convergence to μ) may be interchanged: choosing $n(a) \to \infty$ with $\theta^{\,n(a)} = o(1/a)$ makes the seed-average and the μ-average agree up to $o(1)$. The Lebesgue bound of §9 therefore passes to natural density, and the seeds of scale $< a_\star$ form a density-zero set (as $\sigma \to \infty$, Lemma 6). Hence*

$$\overline{d}\big(\{(a, b) : \text{orbit meets } \mathcal{D}\}\big) \leq \varepsilon.$$

As $\varepsilon > 0$ was arbitrary, the density is 0, proving Theorem 1. Finally, by Lemma 6 every surviving orbit has $a \to \infty$ and never meets $\mathcal{D}$, i.e. never halts. □

10 Remarks on rigor and scope

Every map-specific ingredient is proved: the parity grammar (Lemma 1); the continuum reduction (3); the explicit Markov partition with endpoints $\rho_{k+1} = (3\rho_k + 1)^{-1}$ and accumulation (5) (Lemma 2); the full-branch property (Prop. 1); eventual expansion (Lemma 3); summable variation (Lemma 4); the explicit model-to-exact closeness and the uniform Lasota–Yorke inequality (§7). The analytic engine, Theorem 2, is the classical theory of expanding Markov maps.

The two quantities that entered the argument only through a hypothesis are now given in closed form. Transience is *elementary and unconditional*: $\sigma = a + b$ strictly increases at every step by the explicit amounts of Lemma 6, so $\sigma \to \infty$ on every orbit, and $\Lambda > 0$ because its integrand is pointwise ≥ 0 and positive on I_0 — no invariant measure or numerics are needed for either statement. The halting constant is $C = 6\,h(0)$ (Lemma 7) with $0 < C \leq 6\|h\|_\infty < \infty$; only its finiteness is used, and that is the BV bound on the invariant density. The exact value $C = 6h(0)$ (equivalently the per-renewal drift $\Lambda \approx 0.134$) is a functional of the transfer operator's fixed point h, which for this map — as for every Gauss-type map beyond the classical one — is not elementary; the theorem does not require its evaluation. Finally the interchange of limits in Proposition 3 is made uniform by the spectral gap of Prop. 2: the total-variation convergence rate $M\theta^n$ is uniform in the initial point, so choosing $n(a)$ with $\theta^{n(a)} = o(1/a)$ justifies the double limit. No step now rests on an unproven estimate.

The theorem constrains the generic seed and is silent on any particular one: the certified seed $(6, 13)$ is μ-generic, and deciding whether its own orbit meets $\mathcal{D}$ is Collatz-hard. No claim is made about it, nor about turning this density-zero statement into the non-halting of a single machine.

References

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